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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 · 2026–2027 · Trimester 4 · Section 13 · Team 03

CAR RENTAL & MOBILITY PLATFORM

Review-4 · Results and Impact

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Project Summary



Platform Overview

The Car Rental & Mobility Platform is a polyglot, service-oriented backend for bookings, vehicle catalogs, live tracking, and recommendations.

PostgreSQL, MongoDB, and pgvector hold the data; FastAPI, Node.js, and Spring Boot serve it.

This review presents the results, features, and impact of the platform.

Platform Modules



Bookings & Payments



Vehicle Catalog



Real-Time Tracking



Recommendations



Access & Monitoring

Results 1: Correctness and Security

#	Test	Pass Criterion	Result
R1	Double-booking check	No two bookings for one vehicle and time, even with parallel requests	[Add result]
R2	Saga rollback test	A failed payment releases the vehicle and cancels the booking	[Add result]
R3	API and access tests	Correct responses and validation; 401 or 403 for wrong roles	[Add result]

Results 2: Search and Performance

#	Test	Pass Criterion	Result
R4	Recommendation accuracy	HNSW recall@k stays close to exact search	[Add result]
R5	Query speed	HNSW answers faster than exact search	[Add result]
R6	Load and scaling	Response time stays stable under load; one service scales alone	[Add result]

Technical Competence



Relational Modelling

3NF schema, constraints, indexes, procedures, and triggers.



NoSQL and Vector Design

BSON models, aggregation pipelines, and pgvector with HNSW.



Multi-Framework APIs

FastAPI, Node.js/Express, and Spring Boot working together.



Production Deployment

Docker, Kubernetes, GitHub Actions, and OpenTelemetry.

Technical Competence: CO Mapping

CO	Mapping in the Project
CO1	PostgreSQL schema for bookings and payments, SQL queries, joins, and reports
CO2	SQL vs NoSQL: PostgreSQL for transactions, MongoDB for catalog and reviews, pgvector for search
CO3	FastAPI gateway with REST APIs, JWT/OAuth2, validation, and Swagger / OpenAPI docs
CO4	Backend services: Node.js tracking, Spring Boot inventory and pricing, auth, and admin modules
CO5	Microservice architecture using the Saga pattern and event-driven consistency
CO6	Docker, Kubernetes, GitHub Actions CI/CD, and Prometheus / Grafana monitoring

Innovative Features



Innovation in One Platform

Features that go beyond a single-database rental system



Similar-vehicle suggestions using HNSW search



Saga-based consistency across services



Live availability and tracking with Socket.io



The right database for each workload

Sustainable Features



**Scales only
what is
needed**



**Open-source
tools, no
licence cost**



**Modular, easy-
to-maintain
services**



**Monitoring
finds faults
early**



**Supports
shared
mobility**

Utility to Society



The platform makes vehicle rental easier, safer, and more transparent for everyone involved.



Customers

Easy booking, live availability, and relevant vehicle suggestions.



Agents and Admins

Role-based tools to manage vehicles, bookings, and reports.



Society

Supports affordable, shared mobility and fewer private cars.

Live Demo

Components Shown



FastAPI Gateway
JWT login and Swagger UI



MongoDB
Vehicle catalog and reviews



pgvector
Similar-vehicle search



Node.js + Redis
Live tracking updates



Grafana
Metrics and traces

Demo Walkthrough

1

Log in as a customer through the gateway

2

Browse the vehicle catalog and reviews

3

See similar-vehicle suggestions

4

Book a car; payment completes via the Saga

5

Watch live vehicle tracking update

6

Show metrics and traces in Grafana

Conclusion and Future Scope



The platform shows a polyglot, service-oriented backend for car rental, with consistent bookings, smart recommendations, and live tracking.



Event Streaming

Kafka-based events between services.



Smarter Pricing

ML demand forecasting and dynamic pricing.



Wider Reach

Mobile app, multi-region deployment, and fraud detection.